Financial Mathematics Practice Worksheet

Chapter 11.1 – Interest Periods and Effective Rates

Tasks 1–20, sorted from easiest to hardest (difficulty 1/5 to 5/5). Core formulas from Section 11.1: after t years a principal S_0 earning nominal annual rate r with interest added n times per year grows to $S_0(1 + r/n)^{nt}$; the effective yearly rate is $R = (1 + r/n)^n – 1$. Round dollar amounts to 2 decimal places and rates to 2 decimal places unless otherwise noted. Each task has 5 True/False statements (A–E); the number of true statements per task varies across the whole set (from 1 to 5) rather than being fixed.

This set is built to spread evenly across every exercise type in Section 11.1:

- Converting a nominal rate to an effective yearly rate R, for different n (Tasks 1, 5, 13)
- Multi-year future value $S_0(1+r/n)^{nt}$ for $t \neq 1$ (Tasks 2, 8, 17)
- Solving for time t via logarithms — doubling time, reaching a target (Tasks 6, 12, 20)
- Solving for the required nominal rate r to hit a growth target (Tasks 9, 16)
- Present value — how much had to be invested in the past to reach a value today (Tasks 11, 18)
- Comparing two or more offers to see which is truly better (Tasks 3, 10, 19)
- Converting a periodic (e.g., monthly credit-card) rate to an effective annual rate (Tasks 4, 14)
- Comparing the same nominal rate under several compounding frequencies (Tasks 7, 15)

Task 1. Nominal Rate vs. Effective Rate on a Business Account

(difficulty score 1/5)

A print shop owner deposits $6,000 into a business savings account offering a nominal annual rate of 7.20%, compounded monthly.

Statements (True / False):

a) The monthly periodic interest rate is 0.60%.

b) The effective annual rate is approximately 7.44%.

c) A $6,000 deposit left for exactly one year would grow to $6,446.54.

d) If the bank instead compounded the same nominal rate annually, the effective annual rate would be higher than under monthly compounding.

e) The effective annual rate exceeds the nominal rate by more than 1.00 percentage point.

Solution – Task 1

Given:

P = $6,000

Nominal annual rate r = 7.20% = 0.072

Compounding frequency n = 12 (monthly)

Time = 1 year

Formula(s):

Periodic rate = r/n

$R = (1 + r/n)^n - 1$

FV = P × (1 + R)

Step-by-Step Calculations:

Periodic rate = 0.072/12 = 0.006 = 0.60%.

$R = (1.006)^{12} - 1 \approx 1.074424 - 1 = 0.074424 \approx 7.44\%$.

FV = 6,000 × 1.074424 = $6,446.54.

Annual compounding (n = 1) gives R = nominal rate = 7.20%, which is lower than 7.44%, not higher.

Gap = 7.44% – 7.20% = 0.24 percentage points, which is far less than 1.00 point.

Explanation of Each Statement:

a) TRUE. Periodic rate = 0.072/12 = 0.60%, matching exactly.

b) TRUE. $R = (1.006)^{12} - 1 \approx 7.44\%$, matching exactly.

c) TRUE. FV = 6,000 × 1.074424 = $6,446.54, matching exactly.

d) FALSE. Trap: annual compounding (n = 1) simply reproduces the nominal rate as R, so it would equal 7.20%, which is lower than the monthly-compounded 7.44%, not higher.

e) FALSE. Trap: the actual gap is only 7.44% – 7.20% = 0.24 percentage points, far less than 1.00 percentage point.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: FALSE, e: FALSE

(3 of 5 statements are TRUE) — Covers: R = (1+r/n)^n formula (11.1.2)

Task 2. Multi-Year Growth Under Quarterly Compounding

(difficulty score 1.2/5)

A deposit of $6,000 is put into an account earning interest at the annual rate of 8%, with interest paid quarterly. The owner wants to know how much will be in the account after 6 years.

Statements (True / False):

a) The quarterly periodic rate is 2.00%.

b) The number of quarterly periods over 6 years is 24.

c) The balance after 6 years is approximately $9,860.00.

d) If the deposit were left for only 3 years instead of 6, the future value would be exactly half of the 6-year future value.

e) The total percentage growth of the deposit over the 6 years is more than 65%.

Solution – Task 2

Given:

S_0 = $6,000

Nominal annual rate r = 8% = 0.08

Compounding frequency n = 4 (quarterly)

Time t = 6 years

Formula(s):

Periodic rate = r/n

$S(t) = S_0(1 + r/n)^{nt}$

Step-by-Step Calculations:

Periodic rate = 0.08/4 = 0.02 = 2.00%; nt = 4 × 6 = 24.

S(6) = 6,000 × $(1.02)^{24}$ ≈ 6,000 × 1.608435 = $9,650.61 (not $9,860.00).

Compound growth is exponential, not linear, in time: $(1.02)^{12}$ ≈ 1.268242, so the 3-year value would be about $7,609.45, which is NOT half of $9,650.61 (half would be $4,825.31).

Total growth = (9,650.61 – 6,000)/6,000 ≈ 0.6084 = 60.84%, which is not more than 65%.

Explanation of Each Statement:

a) TRUE. Periodic rate = 0.08/4 = 2.00%, matching exactly.

b) TRUE. nt = 4 × 6 = 24, matching exactly.

c) FALSE. Trap: S(6) = 6,000 × $(1.02)^{24}$ ≈ $9,650.61, not $9,860.00.

d) FALSE. Trap: compound growth is an exponential function of time, not a linear one, so halving the time does not halve the future value. The 3-year balance (≈$7,609.45) is far more than half of the 6-year balance.

e) FALSE. Trap: the actual total growth is (9,650.61 – 6,000)/6,000 ≈ 60.84%, which is less than 65%, not more.

Final Answers — a: TRUE, b: TRUE, c: FALSE, d: FALSE, e: FALSE

(2 of 5 statements are TRUE) — Covers: S(t) = S0(1+r/n)^(nt), multi-year (Example 11.1.1)

Task 3. Which Savings Offer Is Better?

(difficulty score 1.4/5)

A saver has $10,000 and is comparing two one-year term deposits: Offer (i) 6.4% with interest paid quarterly; Offer (ii) 6.5% with interest paid twice a year (semi-annually).

Statements (True / False):

a) The effective annual rate of Offer (i) is approximately 6.55%.

b) The effective annual rate of Offer (ii) is approximately 6.61%.

c) Offer (ii) is the better choice for the saver.

d) Because Offer (i) compounds more frequently, it must have the higher effective rate.

e) Depositing $10,000 for one year, Offer (ii) would produce more than $660 in interest.

Solution – Task 3

Given:

P = $10,000

Offer (i): r = 6.4%, n = 4 (quarterly)

Offer (ii): r = 6.5%, n = 2 (semi-annual)

Time = 1 year

Formula(s):

$R = (1 + r/n)^n – 1$

$FV = P × (1 + R)$

Step-by-Step Calculations:

Offer (i): periodic rate = 0.064/4 = 0.016. $R_i = (1.016)^4 – 1 ≈ 1.065533 – 1 = 0.065533 ≈ 6.55\%$.

Offer (ii): periodic rate = 0.065/2 = 0.0325. $R_{ii} = (1.0325)^2 – 1 = 1.066056 – 1 = 0.066056 ≈ 6.61\%$.

Since R_{ii} (6.61%) > R_i (6.55%), Offer (ii) is the better deal for the saver, despite compounding less often.

$Interest_i$ = 10,000 × 0.065533 = $655.33. $Interest_{ii}$ = 10,000 × 0.066056 = $660.56. Difference = $5.23–$5.24.

Explanation of Each Statement:

a) TRUE. $R_i = (1.016)^4 – 1 ≈ 6.55\%$, matching exactly.

b) TRUE. $R_{ii} = (1.0325)^2 – 1 ≈ 6.61\%$, matching exactly.

c) TRUE. A higher effective annual rate directly means more money for the saver, so Offer (ii) is indeed better.

d) FALSE. Trap: despite compounding less frequently, Offer (ii)'s higher nominal rate (6.5% vs. 6.4%) is enough to give it the higher effective rate (6.61% vs. 6.55%) — frequency alone does not decide which offer is better.

e) TRUE. 10,000 × (0.066056 – 0.065533) ≈ $5.23–$5.24, matching approximately.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: FALSE, e: TRUE

(4 of 5 statements are TRUE) — Covers: Comparing offers via R (Example 11.1.4)

Task 4. Credit Card Interest — Monthly Rate to Effective Annual Rate

(difficulty score 1.6/5)

A retail credit card charges interest on any outstanding balance at a rate of 1.75% per month.

Statements (True / False):

a) The nominal annual rate, quoted as 12 times the monthly rate, is 22.00%.

b) The effective annual rate of interest is approximately 21.75%.

c) An unpaid balance of $2,000 would grow to $2,420.00 after one year of accruing interest this way.

d) The effective annual rate exceeds the nominal annual rate by more than 2.00 percentage points.

e) If the monthly rate were instead 1.50%, the effective annual rate would still exceed 20%.

Solution – Task 4

Given:

Monthly periodic rate = 1.75% = 0.0175

n = 12 (months per year)

Formula(s):

Nominal annual rate = 12 × monthly rate

$R = (1 + \text{monthly rate})^{12} – 1$

Step-by-Step Calculations:

Nominal annual rate = 12 × 1.75% = 21.00% (not 22.00%).

$R = (1.0175)^{12} – 1 \approx 1.231430 – 1 = 0.231430 \approx 23.14\%$ (not 21.75%).

FV of $2,000 unpaid for 1 year = 2,000 × 1.231430 = $2,462.86 (not $2,420.00).

Gap = 23.14% – 21.00% = 2.14 percentage points, which is indeed more than 2.00 percentage points.

At a 1.50% monthly rate: $R = (1.015)^{12} – 1 \approx 1.195625 – 1 = 0.195625 \approx 19.56\%$, which does NOT exceed 20%.

Explanation of Each Statement:

a) FALSE. Trap: 12 × 1.75% = 21.00%, not 22.00%.

b) FALSE. Trap: the true effective annual rate is $(1.0175)^{12} – 1 \approx 23.14\%$, not 21.75%.

c) FALSE. Trap: FV = 2,000 × 1.231430 = $2,462.86, not $2,420.00.

d) TRUE. 23.14% – 21.00% = 2.14 percentage points, which is indeed more than 2.00 percentage points.

e) FALSE. Trap: at a 1.50% monthly rate, the effective annual rate is only about 19.56%, which falls just short of 20%, not above it.

Final Answers — a: FALSE, b: FALSE, c: FALSE, d: TRUE, e: FALSE

(1 of 5 statements are TRUE) — Covers: Periodic rate -> effective annual rate (Exercise 8 style)

Task 5. Quarterly Compounding on a Business Deposit

(difficulty score 1.8/5)

A veterinary clinic deposits $15,000 into a one-year business account earning a nominal annual rate of 5.6%, compounded quarterly.

Statements (True / False):

a) The quarterly periodic rate is 1.40%.

b) The effective annual rate is approximately 5.72%.

c) The balance after one year is approximately $15,857.81.

d) If the same nominal rate were instead compounded monthly, the resulting effective annual rate would be lower than under quarterly compounding.

e) The gap between the EAR and the nominal rate exceeds 0.20 percentage points.

Solution – Task 5

Given:

P = $15,000

Nominal annual rate r = 5.6% = 0.056

Compounding frequency n = 4 (quarterly)

Time = 1 year

Formula(s):

Periodic rate = r/n

$R = (1 + r/n)^n - 1$

FV = P × (1 + R)

Step-by-Step Calculations:

Periodic rate = 0.056/4 = 0.014 = 1.40%.

$R = (1.014)^4 - 1 \approx 1.057187 - 1 = 0.057187 \approx 5.72\%$.

FV = 15,000 × 1.057187 = $15,857.81.

Monthly compounding of the same 5.6% nominal rate gives a slightly higher EAR (≈ 5.746%), not lower — more frequent compounding always raises EAR.

Gap = 5.7187% – 5.60% ≈ 0.12 percentage points, which is less than 0.20 percentage points.

Explanation of Each Statement:

a) TRUE. Periodic rate = 0.056/4 = 1.40%, matching exactly.

b) TRUE. $R = (1.014)^4 - 1 \approx 5.72\%$, matching exactly.

c) TRUE. FV = 15,000 × 1.057187 = $15,857.81, matching exactly.

d) FALSE. Trap: switching to monthly compounding raises the EAR (to roughly 5.75%), since more frequent compounding never lowers the EAR for a fixed nominal rate.

e) FALSE. Trap: the actual gap is only about 0.12 percentage points, which does not exceed 0.20 percentage points.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: FALSE, e: FALSE

(3 of 5 statements are TRUE) — Covers: $R = (1+r/n)^n$ formula (11.1.2)

Task 6. How Long to Double at 7.2% Compounded Monthly?

(difficulty score 2.0/5)

A savings account earns a nominal annual rate of 7.2%, compounded monthly. An investor wants to know how long it will take for a deposit to double in value.

Statements (True / False):

a) The monthly periodic rate is 0.60%.

b) It takes approximately 108 months for the deposit to double.

c) It would take approximately 58 months for the deposit to double, which would be exactly half of the actual doubling time.

d) If the nominal rate were doubled to 14.4%, the resulting effective annual rate would also be exactly double the original effective annual rate.

e) Because $(1 + r/n)^t$ is an increasing function of t, the same logarithmic method used for doubling can be used to find the time needed for any other target growth multiple.

Solution – Task 6

Given:

Nominal annual rate r = 7.2% = 0.072

Compounding frequency n = 12 (monthly)

Target: double the principal

Formula(s):

Periodic rate = r/n

$(1 + r/n)^t$ = target multiple

t = ln(target)/ln(1 + r/n)

Step-by-Step Calculations:

Periodic rate = 0.072/12 = 0.006 = 0.60%.

Solve $(1.006)^t = 2$: t = ln 2 / ln 1.006 ≈ 0.693147/0.0059821 ≈ 115.85 months ≈ 9.65 years (not 108 months).

115.85 months is the correct doubling time, not 58 months (which is not even close to a clean half of it, since doubling time does not simply relate this way to a 'half time').

At 14.4% nominal monthly, $R = (1.012)^{12} - 1 \approx 15.38\%$, but double the original 7.44% EAR would be 14.88%, not 15.38% — doubling the nominal rate does not exactly double the EAR.

The same logarithmic method, t = ln(target)/ln(periodic factor), works for any target multiple, since the exponential growth curve is monotonically increasing — this is a general, valid technique.

Explanation of Each Statement:

a) TRUE. Periodic rate = 0.072/12 = 0.60%, matching exactly.

b) FALSE. Trap: the correct doubling time is t = ln2/ln(1.006) ≈ 115.85 months (about 9.65 years), not 108 months.

c) FALSE. Trap: the true doubling time is about 115.85 months; 58 months is nowhere near it and is not meaningfully 'half' of the correct value in any valid sense.

d) FALSE. Trap: doubling the nominal rate raises the EAR to about 15.38%, but exactly double the original EAR of 7.44% would be 14.88% — the two figures are close but not equal, since EAR is a nonlinear (not proportional) function of the nominal rate.

e) TRUE. Because $(1 + r/n)^t$ is a strictly increasing exponential function of t, solving for t via natural logarithms works for any target multiple, not just doubling — this is exactly the method used in Example 11.1.2 of the textbook.

Final Answers — a: TRUE, b: FALSE, c: FALSE, d: FALSE, e: TRUE

(2 of 5 statements are TRUE) — Covers: Solving for t via logarithms (Example 11.1.2)

Task 7. Comparing Compounding Frequencies at a Fixed Nominal Rate

(difficulty score 2.2/5)

A finance student is asked to compute the effective yearly rate corresponding to a nominal annual interest rate of 15%, with interest added: (a) twice a year (semi-annually); (b) each quarter; (c) each month.

Statements (True / False):

a) The effective rate under semi-annual compounding is approximately 15.56%.

b) The effective rate under quarterly compounding is approximately 15.87%.

c) The effective rate under monthly compounding is approximately 16.08%.

d) Increasing the compounding frequency from semi-annual to quarterly to monthly steadily raises the effective rate.

e) The increase in effective rate from semi-annual to quarterly is smaller than the increase from quarterly to monthly.

Solution – Task 7

Given:

Nominal annual rate $r = 15\% = 0.15$

$n = 2$ (semi-annual), $n = 4$ (quarterly), $n = 12$ (monthly)

Formula(s):

$R = (1 + r/n)^n - 1$

Step-by-Step Calculations:

Semi-annual: $R = (1.075)^2 - 1 = 1.155625 - 1 = 0.155625 \approx 15.56\%$.

Quarterly: $R = (1.0375)^4 - 1 \approx 1.158650 - 1 = 0.158650 \approx 15.87\%$.

Monthly: $R = (1.0125)^{12} - 1 \approx 1.160766 - 1 = 0.160766 \approx 16.08\%$.

Ranking confirms $15.56\% < 15.87\% < 16.08\%$ as compounding frequency increases.

First gap = 15.87% – 15.56% = 0.31 points; second gap = 16.08% – 15.87% = 0.21 points. The first gap is actually LARGER than the second, not smaller — each additional increase in frequency adds progressively less to the EAR.

Explanation of Each Statement:

a) TRUE. $R = (1.075)^2 - 1 = 15.5625\%$, matching exactly.

b) TRUE. $R = (1.0375)^4 - 1 \approx 15.87\%$, matching exactly.

c) TRUE. $R = (1.0125)^{12} - 1 \approx 16.08\%$, matching exactly.

d) TRUE. This textbook fact (Example 11.2.3 referenced in Section 11.1) is confirmed here: for a fixed $r > 0$, R strictly increases as n increases.

e) FALSE. Trap: the gap from semi-annual to quarterly (0.31 points) is actually larger than the gap from quarterly to monthly (0.21 points), not smaller — the marginal benefit of increasing compounding frequency diminishes as n grows, which is why R never grows without bound even as $n \to \infty$.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: TRUE, e: FALSE

(4 of 5 statements are TRUE) — Covers: R increasing in n, multiple frequencies (Exercise 5 style)

Task 8. A Ten-Year Deposit Under Monthly Compounding

(difficulty score 2.4/5)

A grandparent deposits $4,000 into a trust account for a grandchild, earning a nominal annual rate of 6%, compounded monthly, for 10 years.

Statements (True / False):

a) The monthly periodic rate is 0.50%.

b) The number of monthly compounding periods over 10 years, nt, is 120.

c) The balance after 10 years is approximately $7,277.60.

d) The deposit exactly doubles in value over these 10 years.

e) If compounded annually instead, the 10-year future value would exceed the future value obtained under monthly compounding.

Solution – Task 8

Given:

S_0 = $4,000

Nominal annual rate r = 6% = 0.06

Compounding frequency n = 12 (monthly)

Time t = 10 years

Formula(s):

$S(t) = S_0(1 + r/n)^{nt}$

Step-by-Step Calculations:

Periodic rate = 0.06/12 = 0.005 = 0.50%; nt = 12 × 10 = 120.

$S(10) = 4{,}000 \times (1.005)^{120} \approx 4{,}000 \times 1.8194 = \$7{,}277.60$.

Growth factor is 1.8194, not 2.0, so the deposit has NOT quite doubled after 10 years (it would take about 11.6 years to fully double at this rate).

Annual compounding (n = 1): $S(10) = 4{,}000 \times (1.06)^{10} = 4{,}000 \times 1.790847 = \$7{,}163.39$, which is LESS than $7,277.60, not more, since less frequent compounding always yields a smaller future value for the same nominal rate.

Explanation of Each Statement:

a) TRUE. Periodic rate = 0.06/12 = 0.50%, matching exactly.

b) TRUE. nt = 12 × 10 = 120, matching exactly.

c) TRUE. $S(10) = 4{,}000 \times (1.005)^{120} \approx \$7{,}277.60$, matching exactly.

d) FALSE. Trap: the growth factor is 1.8194, so the deposit is close to but has not yet doubled; it would need roughly 11.6 years, not 10, to fully double at this rate.

e) FALSE. Trap: annual compounding produces a SMALLER future value ($7,163.39) than monthly compounding ($7,277.60) for the same nominal rate, since less frequent compounding always yields less growth, not more.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: FALSE, e: FALSE

(3 of 5 statements are TRUE) — Covers: $S(t) = S0(1+r/n)^{(nt)}$, multi-year (Example 11.1.1)

Task 9. What Rate Is Needed to Grow $50,000 to $80,000 in 8 Years?

(difficulty score 2.6/5)

An investment fund wants to grow $50,000 into $80,000 over 8 years, with interest compounded quarterly. The fund manager needs to find the required nominal annual rate.

Statements (True / False):

a) The required nominal annual rate, compounded quarterly, is approximately 5.92%.

b) The corresponding quarterly periodic rate is approximately 1.48%.

c) If the same growth were required in only 4 years instead of 8, the required nominal rate would be lower than in the original 8-year scenario.

d) If the compounding were changed from quarterly to monthly, the required nominal rate would need to be higher than in the original quarterly scenario.

e) Growing from $50,000 to $80,000 represents an increase of more than 65%.

Solution – Task 9

Given:

S_0 = $50,000

Target S(t) = $80,000

n = 4 (quarterly)

t = 8 years, so nt = 32

Formula(s):

$S(t) = S_0(1 + r/n)^{nt}$

$r = n \times [(S(t)/S_0)^{1/nt} - 1]$

Step-by-Step Calculations:

$(1 + r/4)^{32} = 80{,}000/50{,}000 = 1.6$, so $1 + r/4 = (1.6)^{1/32} \approx 1.014796$, giving $r \approx 4 \times 0.014796 \approx 0.05918 \approx$ 5.92%.

Quarterly periodic rate = 5.92%/4 ≈ 1.48%.

A shorter time horizon (4 years instead of 8) to reach the same growth factor of 1.6 requires a HIGHER rate, not a lower one, since less time demands faster growth.

Monthly compounding is more frequent than quarterly, so it needs a LOWER nominal rate, not a higher one, to reach the same 1.6 growth factor over the same 8 years.

Growth = (80,000 – 50,000)/50,000 = 30,000/50,000 = 0.60 = 60.00% exactly, which is not more than 65%.

Explanation of Each Statement:

a) TRUE. Solving $(1 + r/4)^{32} = 1.6$ gives r ≈ 5.92%, matching exactly.

b) TRUE. Quarterly periodic rate = 5.92%/4 ≈ 1.48%, matching exactly.

c) FALSE. Trap: reaching the same 1.6× growth factor in HALF the time (4 years instead of 8) requires a substantially HIGHER rate, not a lower one.

d) FALSE. Trap: because monthly compounding is more frequent than quarterly, it actually needs a LOWER nominal rate than 5.92% to reach the same target — not a higher one.

e) FALSE. Trap: the actual increase is exactly (80,000 – 50,000)/50,000 = 60.00%, which is not more than 65%.

Final Answers — a: TRUE, b: TRUE, c: FALSE, d: FALSE, e: FALSE

(2 of 5 statements are TRUE) — Covers: Solving for the nominal rate r

Task 10. Which Terms Are Better for a Borrower?

(difficulty score 2.8/5)

A borrower is comparing two loan terms: option (a) a nominal annual rate of 10.80%, with interest paid annually; option (b) a nominal annual rate of 10.40%, with interest paid quarterly.

Statements (True / False):

a) Option (a)'s effective annual rate is higher than option (b)'s effective annual rate.

b) Option (b)'s effective annual rate is approximately 10.81%.

c) Because option (b) quotes a lower nominal rate, it must be the cheaper option for the borrower.

d) For the borrower, option (a) is more expensive than option (b).

e) The two options' effective annual rates differ by more than 0.05 percentage points.

Solution – Task 10

Given:

Option (a): $r = 10.80\%$, $n = 1$ (annual)

Option (b): $r = 10.40\%$, $n = 4$ (quarterly)

Time = 1 year

Formula(s):

$R = (1 + r/n)^n - 1$

Step-by-Step Calculations:

Option (a): with $n = 1$, R_a = nominal rate = 10.80% exactly.

Option (b): periodic rate = 0.104/4 = 0.026. $R_b = (1.026)^4 - 1 \approx 1.108127 - 1 = 0.108127 \approx 10.81\%$.

Since R_b (10.81%) is very slightly higher than R_a (10.80%), option (a) is actually marginally cheaper for the borrower, even though its quoted nominal rate is higher.

Difference = 10.8127% – 10.80% ≈ 0.013 percentage points, which is less than 0.05 percentage points.

Explanation of Each Statement:

a) FALSE. Trap: $R_a = 10.80\%$ is actually slightly LOWER than $R_b \approx 10.81\%$, not higher.

b) TRUE. $R_b = (1.026)^4 - 1 \approx 10.81\%$, matching exactly.

c) FALSE. Trap: despite quoting a lower nominal rate, option (b)'s quarterly compounding pushes its effective rate very slightly above option (a)'s, so option (b) is actually marginally more expensive, not cheaper.

d) FALSE. Trap: since 10.80% < 10.81%, option (a) is actually the slightly CHEAPER (less expensive) option for the borrower, not the more expensive one.

e) FALSE. Trap: the actual difference is only about 0.013 percentage points, which is less than 0.05 percentage points, not more.

Final Answers — a: FALSE, b: TRUE, c: FALSE, d: FALSE, e: FALSE

(1 of 5 statements are TRUE) — Covers: Comparing offers, borrower's perspective (Exercise 6 style)

Task 11. How Much Was Deposited 6 Years Ago?

(difficulty score 3.0/5)

A trustee wants to know how much would have needed to be deposited 6 years ago, at a constant annual interest rate of 4.5% (paid once a year), to have exactly $40,000 available today.

Statements (True / False):

a) The growth factor over the 6 years at 4.5% annual compounding is approximately 1.302253.

b) The amount that would need to have been deposited 6 years ago is approximately $30,715.86.

c) This present value is less than $32,000.

d) If the rate had instead been 5.5%, the required present value would be higher than at 4.5%.

e) The total interest earned over the 6 years on this deposit would be approximately $9,284.14.

Solution – Task 11

Given:

Target $S(t)$ = $40,000

$r = 4.5\% = 0.045$

$n = 1$ (annual)

$t = 6$ years

Formula(s):

$S(t) = S_0(1 + r)^t$

$S_0 = S(t)/(1 + r)^t$

Step-by-Step Calculations:

$(1.045)^6 \approx 1.302253$.

$S_0 = 40{,}000/1.302253 \approx \$30{,}715.86$.

$30,715.86 is indeed less than $32,000.

A higher rate (5.5%) means the deposit grows faster, so LESS principal is needed today to reach the same $40,000 in 6 years — the required present value would be lower, not higher.

Interest = 40,000 – 30,715.86 = $9,284.14.

Explanation of Each Statement:

a) TRUE. $(1.045)^6 \approx 1.302253$, matching exactly.

b) TRUE. $S_0 = 40{,}000/1.302253 \approx \$30{,}715.86$, matching exactly.

c) TRUE. $30,715.86 < $32,000, matching exactly.

d) FALSE. Trap: a higher interest rate (5.5%) grows money faster, so a SMALLER deposit today would suffice to reach the same $40,000 target — the required present value would be lower, not higher.

e) TRUE. 40,000 – 30,715.86 = $9,284.14, matching exactly.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: FALSE, e: TRUE

(4 of 5 statements are TRUE) — Covers: Present value / backward calculation (Exercise 7(b) style)

Task 12. How Long to Grow £4,000 to £6,000?

(difficulty score 3.2/5)

A deposit of £4,000 earns a nominal annual rate of 6%, compounded monthly. An investor wants to know how long it will take for the balance to reach £6,000.

Statements (True / False):

a) It takes approximately 81.30 months to reach £6,000.

b) It takes approximately 6.00 years exactly to reach £6,000.

c) It takes approximately 48 months to reach £6,000.

d) Because the deposit needs to grow by a factor of 1.5 rather than double, the required time must be less than half of this account's doubling time.

e) The time required to reach £6,000, found using logarithms, is exactly 100 months.

Solution – Task 12

Given:

S_0 = £4,000

Target $S(t)$ = £6,000

$r = 6\% = 0.06$

$n = 12$ (monthly)

Formula(s):

$(1 + r/n)^t = S(t)/S_0$

$t = \ln(S(t)/S_0) / \ln(1 + r/n)$

Step-by-Step Calculations:

Periodic rate = 0.06/12 = 0.005. Target ratio = 6,000/4,000 = 1.5.

$t = \ln(1.5)/\ln(1.005) \approx 0.405465/0.0049875 \approx 81.30$ months ≈ 6.78 years.

This is neither exactly 6.00 years nor 48 months (4 years) — both are incorrect approximations of the true 81.30-month figure.

Doubling time for this account = $\ln 2/\ln(1.005) \approx 138.99$ months; half of that is ≈ 69.5 months. The actual time to grow by 1.5× is 81.30 months, which is MORE than half the doubling time, not less.

$t = \ln(1.5)/\ln(1.005) \approx 81.30$, not 100.

Explanation of Each Statement:

a) TRUE. $t = \ln(1.5)/\ln(1.005) \approx 81.30$ months ≈ 6.78 years, matching exactly.

b) FALSE. Trap: the correct time is approximately 6.78 years, not exactly 6.00 years.

c) FALSE. Trap: 48 months (4 years) is far short of the correct 81.30 months.

d) FALSE. Trap: the doubling time is about 138.99 months, so half of that is about 69.5 months; the actual 81.30 months needed for a 1.5× increase is MORE than that half, not less.

e) FALSE. Trap: the correct value is approximately 81.30 months, not 100.

Final Answers — a: TRUE, b: FALSE, c: FALSE, d: FALSE, e: FALSE

(1 of 5 statements are TRUE) — Covers: Solving for t via logarithms (Example 11.1.2 / Exercise 4(b) style)

Task 13. Daily Compounding on a Money-Market Deposit

(difficulty score 3.4/5)

A retiree deposits $20,000 into a money-market account earning a nominal annual rate of 4.25%, compounded daily (365-day year), for one year.

Statements (True / False):

a) The daily periodic rate is approximately 0.011644%.

b) The effective annual rate is approximately 4.34%.

c) The balance after one year is approximately $20,868.26.

d) If compounded monthly instead, the effective annual rate would be higher than under daily compounding.

e) The gap between the effective annual rate and the nominal rate exceeds 0.20 percentage points.

Solution – Task 13

Given:

P = $20,000

Nominal annual rate r = 4.25% = 0.0425

Compounding frequency n = 365 (daily)

Time = 1 year

Formula(s):

Periodic rate = r/n

$R = (1 + r/n)^n - 1$

$FV = P \times (1 + R)$

Step-by-Step Calculations:

Periodic rate = 0.0425/365 ≈ 0.00011644 = 0.011644%.

$R = (1 + 0.0425/365)^{365} - 1 \approx 1.043413 - 1 = 0.043413 \approx 4.34\%$.

FV = 20,000 × 1.043413 = $20,868.26.

Monthly compounding of 4.25% nominal gives R ≈ 4.33%, which is slightly LOWER than the daily EAR of 4.34%, not higher — daily compounding is more frequent than monthly.

Gap = 4.34% – 4.25% = 0.09 percentage points, which does not exceed 0.20 percentage points.

Explanation of Each Statement:

a) TRUE. Periodic rate = 0.0425/365 ≈ 0.011644%, matching exactly.

b) TRUE. R ≈ 4.34%, matching exactly.

c) TRUE. FV = 20,000 × 1.043413 = $20,868.26, matching exactly.

d) FALSE. Trap: monthly compounding is less frequent than daily, so it produces a slightly LOWER EAR (≈4.33%), not a higher one.

e) FALSE. Trap: the actual gap is only about 0.09 percentage points, which does not exceed 0.20 percentage points.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: FALSE, e: FALSE

(3 of 5 statements are TRUE) — Covers: $R = (1+r/n)^n$ formula, daily compounding

Task 14. A Store Card's True Annual Cost

(difficulty score 3.6/5)

A retail store credit card charges interest on unpaid balances at a rate of 1.9% per month.

Statements (True / False):

a) The nominal annual rate, quoted as 12 times the monthly rate, is 22.80%.

b) The effective annual rate of interest is approximately 25.34%.

c) The effective annual rate is approximately 22.80%, the same as the nominal annual rate.

d) A $3,000 unpaid balance would grow to $3,684.00 after one year.

e) The effective annual rate exceeds the nominal annual rate by more than 3.00 percentage points.

Solution – Task 14

Given:

Monthly periodic rate = 1.9% = 0.019

n = 12 (months per year)

Formula(s):

Nominal annual rate = 12 × monthly rate

$R = (1 + \text{monthly rate})^{12} - 1$

Step-by-Step Calculations:

Nominal annual rate = 12 × 1.9% = 22.80%.

$R = (1.019)^{12} - 1 \approx 1.253408 - 1 = 0.253408 \approx 25.34\%$.

FV of $3,000 unpaid for 1 year = 3,000 × 1.253408 = $3,760.22 (not $3,684.00).

Gap = 25.34% – 22.80% ≈ 2.54 percentage points, which does not exceed 3.00 percentage points.

Explanation of Each Statement:

a) TRUE. 12 × 1.9% = 22.80%, matching exactly.

b) TRUE. $R = (1.019)^{12} - 1 \approx 25.34\%$, matching exactly.

c) FALSE. Trap: the effective annual rate (25.34%) is meaningfully higher than the nominal annual rate (22.80%), not the same — whenever n > 1, R always exceeds the nominal rate.

d) FALSE. Trap: FV = 3,000 × 1.253408 = $3,760.22, not $3,684.00.

e) FALSE. Trap: the actual gap is about 2.54 percentage points, which is less than 3.00 percentage points, not more.

Final Answers — a: TRUE, b: TRUE, c: FALSE, d: FALSE, e: FALSE

(2 of 5 statements are TRUE) — Covers: Periodic rate -> effective annual rate (Exercise 8 style)

Task 15. Effective Rates for a 10% Nominal Rate Under Three Frequencies

(difficulty score 3.8/5)

A bank wants to publish the effective yearly rate corresponding to a nominal annual rate of 10%, with interest added: (a) twice a year; (b) each quarter; (c) each month.

Statements (True / False):

a) The effective rate under semi-annual compounding is approximately 10.25%.

b) The effective rate under quarterly compounding is approximately 10.38%.

c) The effective rate under monthly compounding is approximately 10.47%.

d) Increasing the compounding frequency from semi-annual to quarterly to monthly steadily raises the effective rate.

e) The jump in effective rate from semi-annual to quarterly is smaller than the jump from quarterly to monthly.

Solution – Task 15

Given:

Nominal annual rate $r = 10\% = 0.10$

$n = 2$ (semi-annual), $n = 4$ (quarterly), $n = 12$ (monthly)

Formula(s):

$R = (1 + r/n)^n – 1$

Step-by-Step Calculations:

Semi-annual: $R = (1.05)^2 – 1 = 1.1025 – 1 = 0.1025 = 10.25\%$.

Quarterly: $R = (1.025)^4 – 1 \approx 1.103813 – 1 = 0.103813 \approx 10.38\%$.

Monthly: $R = (1.0083333)^{12} – 1 \approx 1.104713 – 1 = 0.104713 \approx 10.47\%$.

Confirms increasing R with increasing frequency: $10.25\% < 10.38\% < 10.47\%$.

First jump = 10.38% – 10.25% = 0.13 points; second jump = 10.47% – 10.38% = 0.09 points — the first jump is LARGER than the second, not smaller.

Explanation of Each Statement:

a) TRUE. $R = (1.05)^2 – 1 = 10.25\%$, matching exactly.

b) TRUE. $R = (1.025)^4 – 1 \approx 10.38\%$, matching exactly.

c) TRUE. $R = (1.0083333)^{12} – 1 \approx 10.47\%$, matching exactly.

d) TRUE. This matches the textbook result that R strictly increases in n for a fixed $r > 0$.

e) FALSE. Trap: the first jump (0.13 points) is larger than the second jump (0.09 points), not smaller — the marginal gain from increasing compounding frequency diminishes as n grows.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: TRUE, e: FALSE

(4 of 5 statements are TRUE) — Covers: R increasing in n, multiple frequencies (Exercise 5 style)

Task 16. What Growth Rate Would Multiply GDP by 50 in 80 Years?

(difficulty score 4.0/5)

An economist asks what constant annual percentage growth rate would be needed for a country's GDP to become 50 times as large after 80 years.

Statements (True / False):

a) The required annual growth rate is approximately 5.01%.

b) The required annual growth rate is approximately 6.25%.

c) Since 50 is half of 100, the required rate for 50× growth would be exactly half of the rate needed for 100× growth over the same 80 years.

d) At a growth rate of 5.01% per year, GDP would multiply by exactly 100 after 160 years.

e) Achieving 50× growth in only 40 years would require an annual rate lower than 5.01%.

Solution – Task 16

Given:

Target growth factor = 50

$t = 80$ years

Formula(s):

$(1 + r)^t$ = target factor

$r = (\text{target factor})^{1/t} - 1$

Step-by-Step Calculations:

$(1 + r)^{80} = 50$, so $r = 50^{1/80} - 1 \approx 1.050115 - 1 = 0.050115 \approx 5.01\%$.

This is not 6.25%.

Growth rates are found via a root (an exponential relationship), not a simple linear halving of the target multiple, so a rate for 50× growth is not simply half the rate for 100× growth.

At $r = 5.01\%$, after 160 years the growth factor is $(1.050115)^{160} = (50)^2 = 2{,}500$, not 100.

Achieving the same 50× growth in only 40 years (half the time) requires SOLVING $r = 50^{1/40} - 1 \approx 9.65\%$, which is HIGHER than 5.01%, not lower — less time always demands a faster rate for the same target multiple.

Explanation of Each Statement:

a) TRUE. $r = 50^{1/80} - 1 \approx 5.01\%$, matching exactly.

b) FALSE. Trap: the correct rate is approximately 5.01%, not 6.25%.

c) FALSE. Trap: because the relationship between growth factor and rate is exponential (via a root), rates do not scale linearly with the target multiple; halving the multiple does not halve the required rate.

d) FALSE. Trap: at $r = 5.01\%$, doubling the time to 160 years squares the growth factor ($50^2 = 2{,}500$), not just doubles it to 100.

e) FALSE. Trap: reaching the same 50× growth in half the time (40 years instead of 80) requires a substantially HIGHER rate (≈9.65%), not a lower one.

Final Answers — a: TRUE, b: FALSE, c: FALSE, d: FALSE, e: FALSE

(1 of 5 statements are TRUE) — Covers: Solving for the growth rate r (Exercise 3 style)

Task 17. Saving for a $25,000 Tuition Bill in 7 Years

(difficulty score 4.2/5)

A parent needs exactly $25,000 in 7 years for a tuition bill and is deciding how much to deposit today into one of two accounts: Account X, nominal 5.00% compounded monthly; Account Y, nominal 5.10% compounded quarterly.

Statements (True / False):

a) The amount needed today in Account X to reach $25,000 in 7 years is approximately $17,629.99.

b) The amount needed today in Account Y to reach $25,000 in 7 years is approximately $17,534.28.

c) Account X requires a smaller upfront deposit than Account Y to reach the same $25,000 target.

d) Account Y's effective annual rate is higher than Account X's.

e) Because Account X compounds more frequently, Account X must always require the smaller upfront deposit for any future target and any time horizon.

Solution – Task 17

Given:

Target = $25,000 in t = 7 years

Account X: r = 5.00%, n = 12 (monthly)

Account Y: r = 5.10%, n = 4 (quarterly)

Formula(s):

$S_0 = \text{Target} / (1 + r/n)^{nt}$

$R = (1 + r/n)^n – 1$

Step-by-Step Calculations:

Account X: $(1 + 0.05/12)^{84} \approx 1.418038$. $S_{0,X} = 25{,}000/1.418038 \approx \$17{,}629.99$.

Account Y: $(1 + 0.051/4)^{28} \approx 1.425964$. $S_{0,Y} = 25{,}000/1.425964 \approx \$17{,}534.28$.

Since $S_{0,Y} < S_{0,X}$, Account Y actually requires the SMALLER upfront deposit, not Account X.

$R_X = (1 + 0.05/12)^{12} – 1 \approx 5.12\%$. $R_Y = (1.01275)^4 – 1 \approx 5.20\%$. Indeed $R_Y > R_X$, consistent with Account Y needing less principal.

This scenario itself is a counterexample to statement (e): despite compounding more often, Account X needs the LARGER deposit here, because Account Y's higher nominal rate wins out.

Explanation of Each Statement:

a) TRUE. $S_{0,X} = 25{,}000/1.418038 \approx \$17{,}629.99$, matching exactly.

b) TRUE. $S_{0,Y} = 25{,}000/1.425964 \approx \$17{,}534.28$, matching exactly.

c) FALSE. Trap: Account Y actually requires the smaller upfront deposit ($17,534.28 vs. $17,629.99), not Account X.

d) TRUE. $R_Y \approx 5.20\%$ is indeed higher than $R_X \approx 5.12\%$, matching exactly.

e) FALSE. Trap: this task is itself a counterexample — Account X compounds more frequently but still requires the LARGER deposit, because Account Y's higher nominal rate more than compensates for its lower compounding frequency. Frequency alone never determines which account is better; the nominal rate matters too.

Final Answers — a: TRUE, b: TRUE, c: FALSE, d: TRUE, e: FALSE

(3 of 5 statements are TRUE) — Covers: S(t) formula used for present value + decision-making

Task 18. How Much Was Invested 9 Years Ago?

(difficulty score 4.4/5)

A trustee needs to know how much would have had to be invested 9 years ago, compounded quarterly at a nominal annual rate of 4.4%, to have exactly $60,000 available today.

Statements (True / False):

a) The growth factor over the 9 years is approximately 1.4827.

b) The amount that needed to be invested 9 years ago is approximately $40,467.83.

c) This present value is more than $45,000.

d) If the rate had instead been 5.0% nominal, the required present value would be higher than at 4.4%.

e) The implied total interest earned over the 9 years on this investment is more than $20,000.

Solution – Task 18

Given:

Target = $60,000 today

$r = 4.4\% = 0.044$

$n = 4$ (quarterly)

$t = 9$ years, so $nt = 36$

Formula(s):

$S_0 = \text{Target} / (1 + r/n)^{nt}$

Step-by-Step Calculations:

Periodic rate = 0.044/4 = 0.011. $(1.011)^{36} \approx 1.482658$.

$S_0 = 60{,}000/1.482658 \approx \$40{,}467.83$.

$40,467.83 is less than $45,000, not more.

A higher rate (5.0%) grows money faster, so a SMALLER amount invested 9 years ago would reach $60,000 today — the required present value would be lower, not higher.

Interest = 60,000 – 40,467.83 = $19,532.17, which does not exceed $20,000.

Explanation of Each Statement:

a) TRUE. $(1.011)^{36} \approx 1.4827$, matching exactly.

b) TRUE. $S_0 = 60{,}000/1.482658 \approx \$40{,}467.83$, matching exactly.

c) FALSE. Trap: $40,467.83 is less than $45,000, not more.

d) FALSE. Trap: a higher rate (5.0%) means less principal is needed today to reach the same $60,000 — the present value would be LOWER, not higher.

e) FALSE. Trap: the implied interest is 60,000 – 40,467.83 = $19,532.17, which is just under $20,000, not more.

Final Answers — a: TRUE, b: TRUE, c: FALSE, d: FALSE, e: FALSE

(2 of 5 statements are TRUE) — Covers: Present value / backward calculation (Exercise 7(b) style)

Task 19. Three CDs That Are Closer Than They Look

(difficulty score 4.7/5)

A saver is comparing three one-year certificates of deposit for a $20,000 deposit: CD1: nominal 6.30%, compounded monthly. CD2: nominal 6.40%, compounded quarterly. CD3: nominal 6.45%, compounded semi-annually.

Statements (True / False):

a) CD1's effective annual rate is approximately 6.49%.

b) CD2's effective annual rate is approximately 6.55%.

c) CD3's effective annual rate is approximately 6.55%, essentially the same as CD2's.

d) CD1 has both the lowest nominal rate and the lowest effective annual rate of the three CDs.

e) On the $20,000 deposit, choosing CD2 over CD1 would earn approximately $13.61 more in interest over the year.

Solution – Task 19

Given:

P = $20,000

CD1: r = 6.30%, n = 12 (monthly)

CD2: r = 6.40%, n = 4 (quarterly)

CD3: r = 6.45%, n = 2 (semi-annual)

Time = 1 year

Formula(s):

$R = (1 + r/n)^n - 1$

Interest = P × R

Step-by-Step Calculations:

CD1: periodic rate = 0.063/12 = 0.00525. $R_1 = (1.00525)^{12} - 1 \approx 1.064852 - 1 = 0.064852 \approx 6.49\%$.

CD2: periodic rate = 0.064/4 = 0.016. $R_2 = (1.016)^4 - 1 \approx 1.065533 - 1 = 0.065533 \approx 6.55\%$.

CD3: periodic rate = 0.0645/2 = 0.03225. $R_3 = (1.03225)^2 - 1 \approx 1.065540 - 1 = 0.065540 \approx 6.55\%$, essentially tied with CD2 (difference under 0.01 points).

CD1 has the lowest nominal rate (6.30%) and the lowest EAR (6.49%) of the three.

$\text{Interest}_1 = 20{,}000 \times 0.064852 = \$1{,}297.04$. $\text{Interest}_2 = 20{,}000 \times 0.065533 = \$1{,}310.65$. Difference = $13.61.

Explanation of Each Statement:

a) TRUE. $R_1 = (1.00525)^{12} - 1 \approx 6.49\%$, matching exactly.

b) TRUE. $R_2 = (1.016)^4 - 1 \approx 6.55\%$, matching exactly.

c) TRUE. $R_3 \approx 6.55\%$, essentially matching CD2's rate even though CD3 compounds half as often, because CD3's slightly higher nominal rate makes up the difference.

d) TRUE. CD1 has both the lowest quoted nominal rate and, consistent with that, the lowest effective annual rate of the three — no inversion occurs among these three CDs.

e) TRUE. $20{,}000 \times (0.065533 - 0.064852) \approx \$13.61$, matching approximately.

Final Answers — a: TRUE, b: TRUE, c: TRUE, d: TRUE, e: TRUE

(5 of 5 statements are TRUE) — Covers: Comparing offers across three compounding frequencies

Task 20. Which Account Reaches $22,000 First?

(difficulty score 5.0/5)

A family has $15,000 today and wants to know which of two accounts reaches $22,000 sooner: Account M, nominal 6.00% compounded monthly; Account Q, nominal 6.15% compounded quarterly.

Statements (True / False):

a) It would take approximately 76.8 months for Account M to grow $15,000 to $22,000.

b) It would take Account Q the same amount of time as Account M to reach the same target.

c) Because Account Q compounds less frequently, it must take longer than Account M to reach $22,000.

d) Account M's effective annual rate is higher than Account Q's.

e) If both accounts instead needed to reach $30,000, each account would take exactly twice as long as it took to reach $22,000.

Solution – Task 20

Given:

S_0 = $15,000, target = $22,000

Account M: r = 6.00%, n = 12 (monthly)

Account Q: r = 6.15%, n = 4 (quarterly)

Formula(s):

$t = \ln(\text{target}/S_0) / \ln(1 + r/n)$, in periods

$R = (1 + r/n)^n - 1$

Step-by-Step Calculations:

Target ratio = 22,000/15,000 ≈ 1.466667.

Account M: periodic rate = 0.005. t = ln(1.466667)/ln(1.005) ≈ 0.382996/0.0049875 ≈ 76.81 months ≈ 6.40 years.

Account Q: periodic rate = 0.015375. t = ln(1.466667)/ln(1.015375) ≈ 0.382996/0.0152580 ≈ 25.10 quarters ≈ 6.275 years, which is FASTER than Account M's 6.40 years, not identical and not slower.

$R_M = (1.005)^{12} - 1 \approx 6.17\%$. $R_Q = (1.015375)^4 - 1 \approx 6.29\%$. So $R_Q > R_M$, consistent with Q reaching the target faster.

The time to reach a 1.4667× target and the time to reach a 2× target (doubling) are governed by different logarithms (ln 1.4667 vs. ln 2), so the time to double is NOT simply twice the time to reach 1.4667× growth.

Explanation of Each Statement:

a) TRUE. t = ln(1.466667)/ln(1.005) ≈ 76.8 months ≈ 6.40 years, matching exactly.

b) FALSE. Trap: Account Q actually reaches the target faster, in about 6.275 years, not the same 6.40 years as Account M.

c) FALSE. Trap: despite compounding less frequently, Account Q's higher nominal rate (6.15% vs. 6.00%) lets it reach the target faster (≈6.275 years) than Account M (≈6.40 years), not slower.

d) FALSE. Trap: R_Q (≈6.29%) is actually higher than R_M (≈6.17%), not the other way around.

e) FALSE. Trap: the time to reach a different target multiple (2× instead of 1.4667×) is governed by a different logarithm (ln 2 vs. ln 1.4667) and is not simply double the original time; each target multiple requires its own separate calculation.

Final Answers — a: TRUE, b: FALSE, c: FALSE, d: FALSE, e: FALSE

(1 of 5 statements are TRUE) — Covers: Capstone: solving for t + comparing two accounts